

D500 LiDAR Kit

From Waveshare Wiki

Jump to: navigation, search

Overview

Introduction

STL-19P is mainly composed of a laser ranging core, a wireless transmission unit, an angle measuring unit, a motor driving unit, and a mechanical case. The STL-19P ranging core adopts DTOF technology to measure 5000 times each second. When it works, STL-19P emits the infrared laser forward, and the laser is reflected in the single photon receiving unit after encountering the target object. Thus, we get both the time of laser emitting and receiving, the gap between them is the time of flight. We can calculate the distance by dividing the flight time by the speed of light. After receiving distance data, STL-19P will combine them with angel value from the angle measuring unit to comprise the points cloud data, then transmit the points cloud data to an external interface via wireless communication. Meanwhile, the external interface provides PWM to allow the motor driving unit to drive the motor. After the external control unit gets the rotational speed, it will reach the specified speed through PID algorithm closed-loop control to ensure STL-19P works stably.

Feature

- Based on DTOF (Time of Flight) laser ranging technology
- Support UART communication
- Measuring radius 12m
- 60K Lux anti-glare interference
- 5000Hz measurement frequency
- 360° scanning range
- Laser Safety Meets FDA Class 1 Standards



- Compact appearance
- 10000+ hours of service life

Parameter

Typical measuring range	0.03~12m
Sampling frequency	5000 Hz
Scanning frequency	10 Hz
Angular resolution	≤0.72°
Mechanical size	38.59*38.59*33.50mm
Communication interface	UART @ 230400bps
Supply voltage	5V
Working current	290mA
Power consumption	1.45W
Operating temperature range	-10°C~45°C
Scanning range	360°
Ranging accuracy	10mm (300mm<measurement distance≤500mm) 20mm (500mm<measurement distance≤2000mm) 30mm (2000mm<measurement distance≤12000mm)

Application

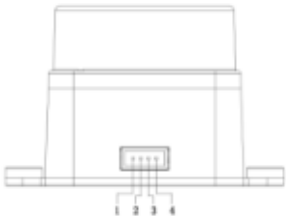
- Education and Research
- Robot obstacle avoidance
- Measurement and Inspection
- Smart Gesture Control
- 2D Gesture Recognition
- Autonomous navigation
- Map reconstruction
- Navigation and positioning

How to use

Communication

STL-19P adopts a ZH1.5T-4P1.5mm standard socket to connect with the external system for power supply, rotation control, and data output. The specific interface definition and parameter requirements are as follows:

No	Function	Type	Description	Minimum	Typical values	Maximum
1	Tx	Output	Lidar Data Output	0V	3.3V	3.5V
2	PWM	Input	Motor Control Signal	0V	-	3.6V
3	GND	Power	Power negative	-	0V	-
4	P5V	Power	Power positive	4.5V	5V	5.5V



(/wiki/File:Ld19.png)

STL-19P has a continuously variable motor driver inside, which supports internal speed control and external speed control. When the PWM pin is not connected or connected to high resistance, the Lidar will use internal speed control, and the default speed is 10Hz. When the PWM pin connects to a square wave signal, the Lidar will use external speed control, and the start, stop and speed of the motor can be controlled by the duty cycle of the PWM signal. Because of the individual differences of each motor driver, the actual speed may be different when the duty cycle is set as a typical value. If need to accurately control the motor speed, you should do closed-loop control via the rotation speed information from the receiving data. The data communication of STL-19P uses a standard asynchronous serial port(UART) in one way, and the transmission parameters are shown below:

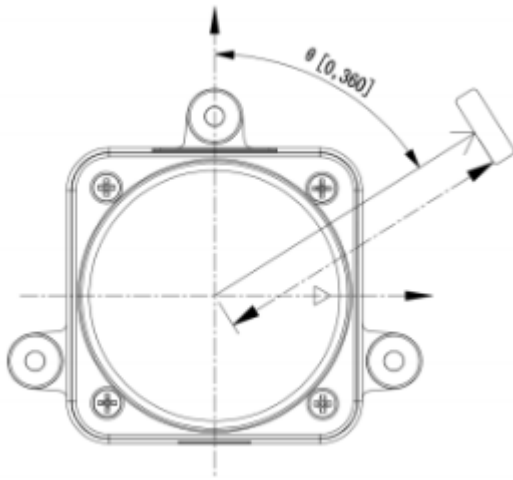
Baud Rate	Data Bits	Stop Bits	Parity	Flow Control
230400	8 Bits	1	No	No

STL-19P adopts one-way communication. After working stably, it begins to send measuring data without any instruction.

Coordinate System Definition

STL-19P uses the left-handed coordinate system, the front side of the sensor is defined as the x-axis (namely the zero direction), the rotation center is the coordinate origin, and the rotation angle increases along the clockwise direction.

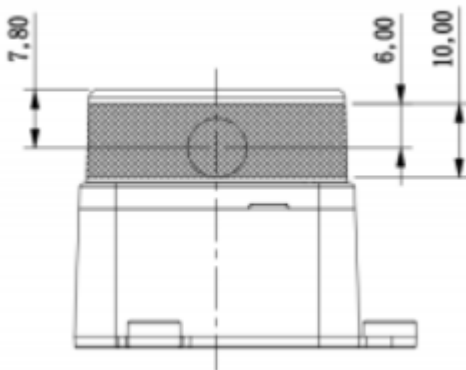
Please refer to the below picture:



(/wiki/File:DTOF-Coordinate.png)

Optical Windows and Mechanical Dimensions

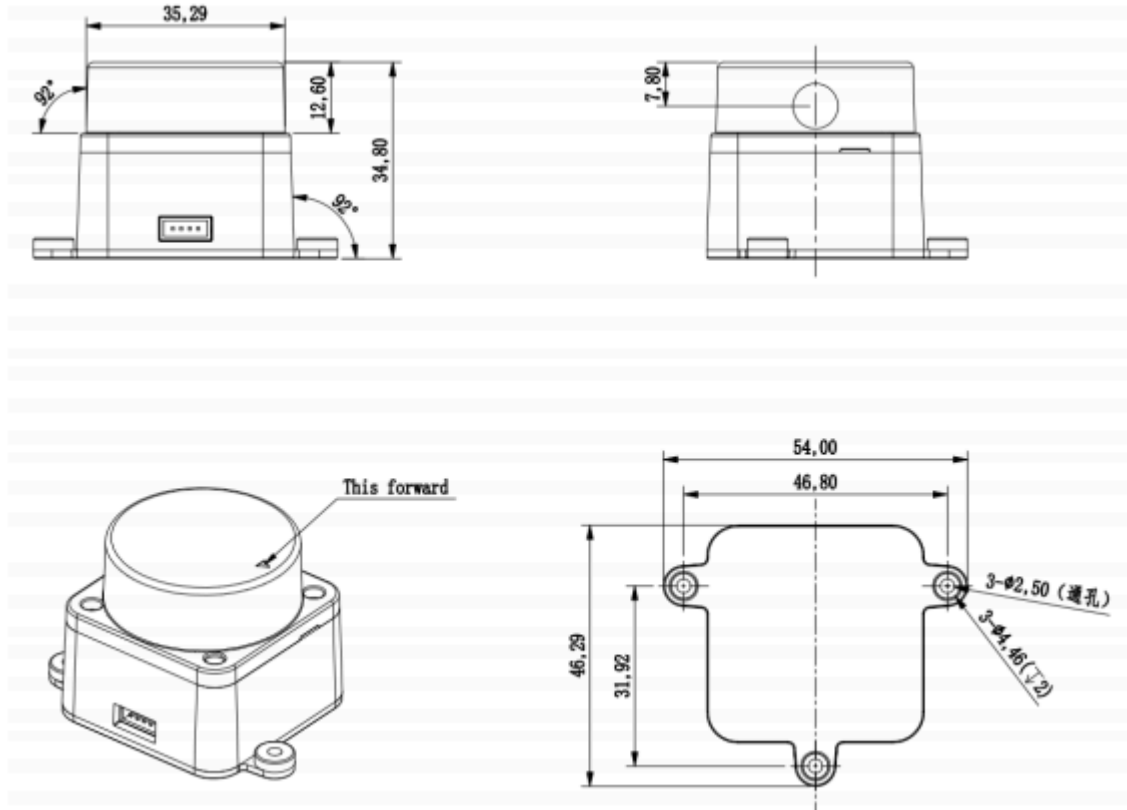
The laser emission and reception of the STL-19P measurement unit will require an optical window that should be structurally sheltered. Any shielding on the outside of the window will affect the measurement performance of the STL-19P to some extent. The following figure shows the dimensions of the optical window (unit: mm):



(/wiki/File:DTOF-Coordinate2.png)

The shaded part in the above picture is the optical window. Please do not cover it. The remaining installation dimensions (Unit: mm) are shown below:

Tolerance: $\pm 0.2\text{mm}$



(/wiki/File:DTOF-

Coordinate3.png)

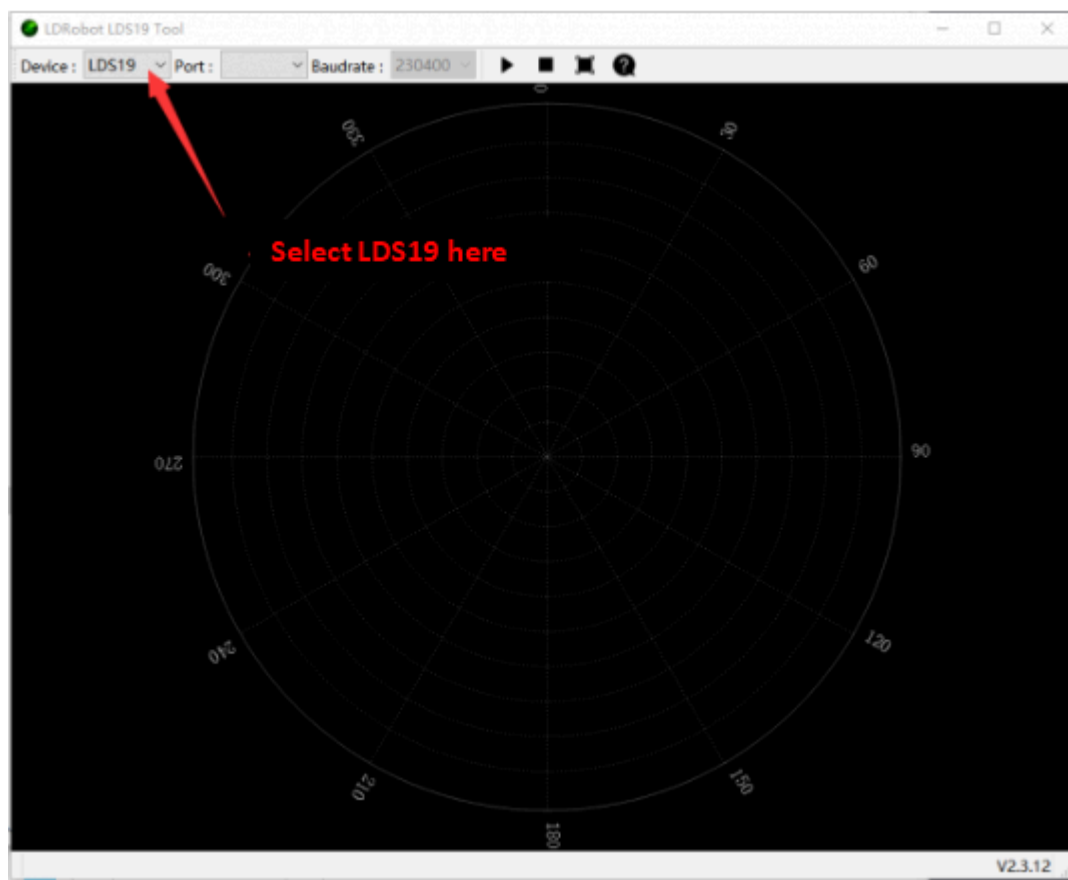
Software

- Ld_desktop_V2.3.12.zip (https://files.waveshare.com/upload/a/a5/Ld_desktop_V2.3.12.zip)
- CP210x_Universal_Windows_Driver.zip (https://files.waveshare.com/upload/6/63/CP210x_Universal_Windows_Driver.zip)
- The Lidar is connected to the adapter board, and then connected to the computer through USB.
- Download Ld_desktop_V2.3.12.zip and unzip it.
- Run Ld_desktop.exe in the compressed package.

bearer	2022/4/12 9:50	文件夹	
iconengines	2022/4/12 9:50	文件夹	
imageformats	2022/4/12 9:50	文件夹	
platforms	2022/4/12 9:50	文件夹	
PointInformationDistanceTable	2022/4/12 9:50	文件夹	
styles	2022/4/12 9:50	文件夹	
translations	2022/4/12 9:50	文件夹	
D3Dcompiler_47.dll	2021/7/2 14:35	应用程序扩展	4,077 KB
DISTANCE.ini	2021/12/29 19:41	配置设置	1 KB
ld_desktop.exe	2022/3/10 14:40	应用程序	183 KB
libEGL.dll	2021/7/2 14:36	应用程序扩展	24 KB
libGLESV2.dll	2021/7/2 14:36	应用程序扩展	3,496 KB
LiDARAdjust_LD06.ini	2022/2/11 13:51	配置设置	1 KB
LidarToolUniform.ini	2022/4/12 11:01	配置设置	1 KB
lrfplayer.ini	2022/2/11 15:26	配置设置	1 KB
opengl32sw.dll	2021/7/2 14:36	应用程序扩展	20,433 KB
Qt5Core.dll	2021/7/2 14:36	应用程序扩展	6,021 KB
Qt5Gui.dll	2021/7/2 14:36	应用程序扩展	6,364 KB
Qt5Network.dll	2021/7/2 14:36	应用程序扩展	1,297 KB
Qt5SerialPort.dll	2021/7/2 14:35	应用程序扩展	85 KB
Qt5Svg.dll	2021/7/2 14:36	应用程序扩展	331 KB
Qt5Widgets.dll	2021/7/2 14:36	应用程序扩展	5,462 KB

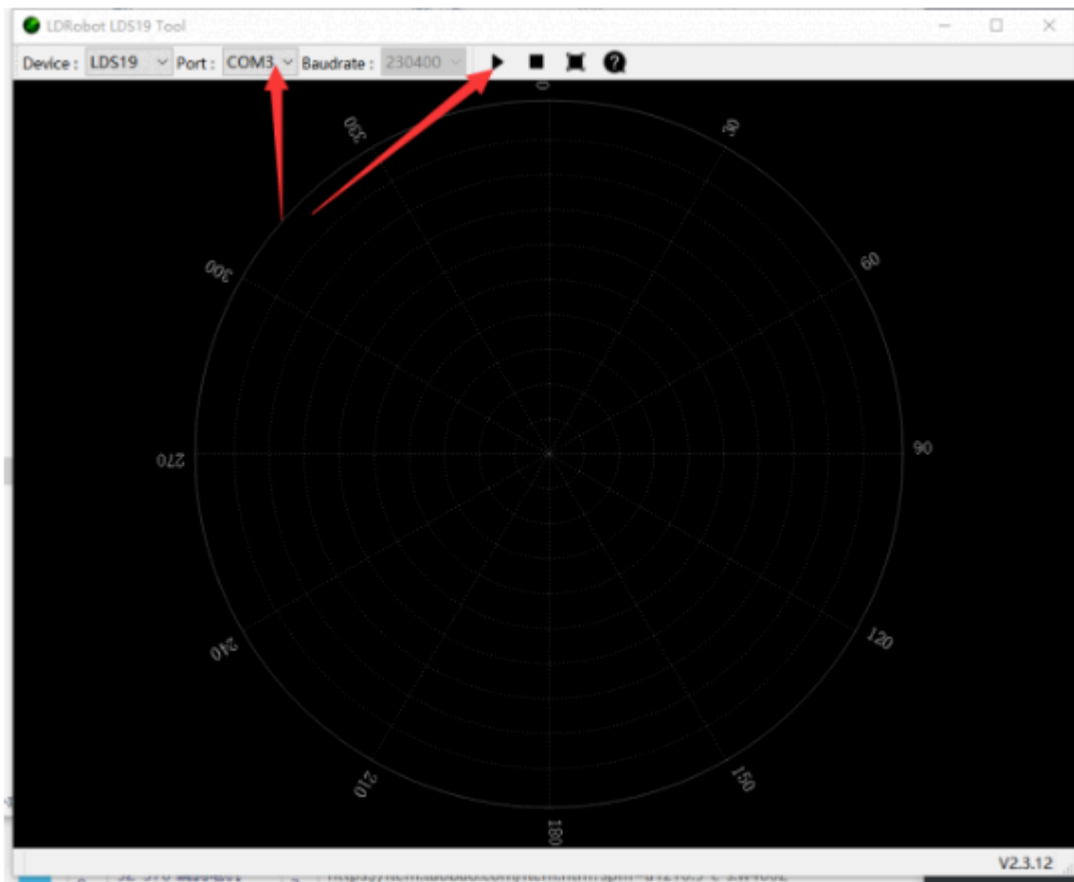
(/wiki/File:Software7.png)

- Select Device LDS19.



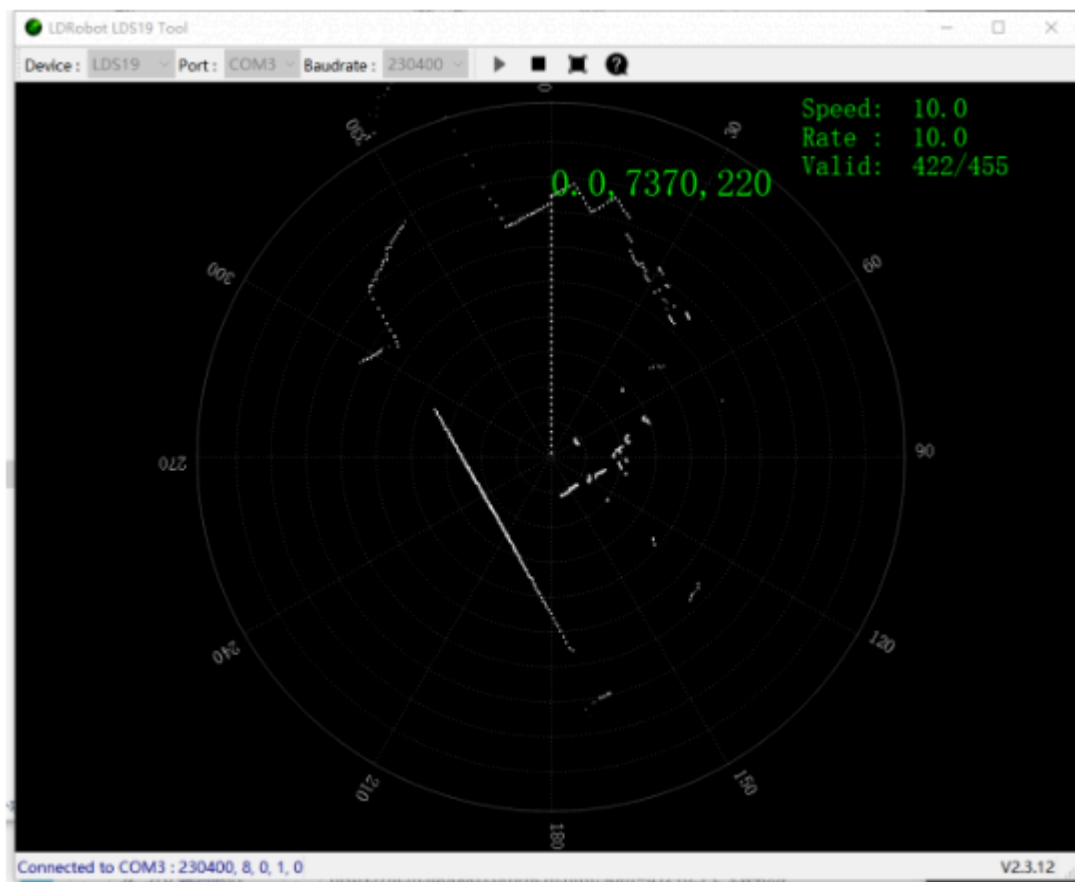
(/wiki/File:LDS19.png)

- Select the COM port. If there is no COM port on the Lidar (CP2102 on the adapter board), you need to download CP210x_Universal_Windows_Driver.zip and install the driver.



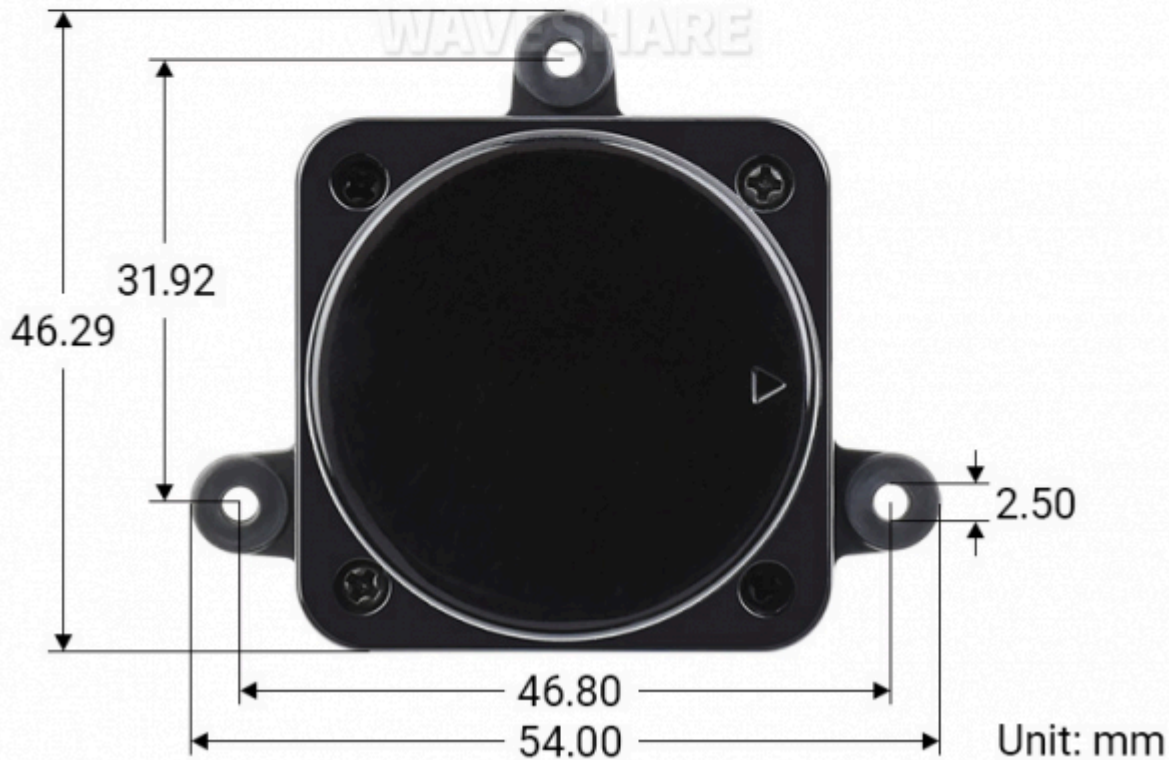
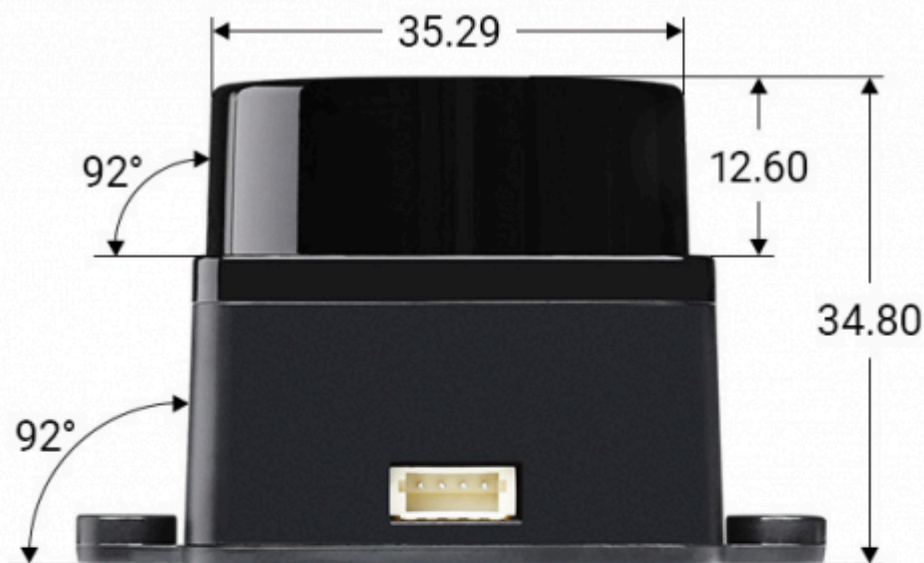
(/wiki/File:COM3.png)

- After clicking the triangle icon above, you can see the scanning results of the radar on the computer.



(/wiki/File:Triangle.png)

Dimensions



(/wiki/File:D500_LiDAR_Kit_Size00.png)

Resource

Document

- DTOF LIDAR STL-19P Specification (<https://files.waveshare.com/wiki/D500-LiDAR-K>)

SDK

- SDK (<https://files.waveshare.com/upload/9/99/Ldrobot-lidar-ros2.zip>)

Software

- Ld_desktop_V2.3.12 (https://files.waveshare.com/upload/a/a5/Ld_desktop_V2.3.12.zip)
- CP210x_Universal Windows Driver (https://files.waveshare.com/upload/6/63/CP210x_Universal_Windows_Driver.zip)

3D Model

- STEP Model (<https://files.waveshare.com/upload/d/d2/DTOF-LIDAR-LD19-STEP.zip>)
- STL Model (https://files.waveshare.com/upload/1/1d/DTOF-LIDAR-LD19-STL_%281%29.zip)

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.

Working Time: 9 AM - 6 PM GMT+8 (Monday to Friday)

Submit Now ()